

Navya Abrol

Robotics Developer | Autonomous Systems | Sensor Fusion
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EDUCATION

Thapar Institute of Engineering & Technology
Bachelor of Technology, Mechatronics

August 2023 – May 2027

EXPERIENCE

Research Intern

February 2026 – Present

Thapar Institute of Engineering & Technology

- Designing and developing an AI-driven ROS2 architecture with integrated perception and path planning for autonomous solar cleaning systems

Research Intern (SPFA – 2025)

June 2025 – August 2025

Indian Institute of Technology, Delhi

- Worked on Static Analysis of Multi-D.o.F. Planar Mechanisms Using Kinematic Coefficients. — Certificate

Exploration Intern

June 2024 – August 2024

Enord, IIITD, Okhla, Delhi NCR

- Contributed to user manual documentation, supported hardware assembly and testing, and proposed process improvements through 6S methodology research. — Certificate

SKILLS

C/C++, Python | ROS2(Humble/Jazzy), Nav2, NVIDIA Isaac, SLAMs, VLMs, EKF, Sensor Fusion, Path Planning | Creo Parametric, MATLAB & simulink | Linux, Git | Raspberry Pi, ESP32, Teensy, UART/I2C, PWM-based Motor Control, Embedded Integration | OpenCV, PyTorch, NumPy, Pandas, Scikit-learn | Robotics Systems (Perception, Navigation, Human - Robot Interaction)

PROJECTS

Audio-Visual-Thermal Fusion for Autonomous Rescue Robot

February 2026 – Ongoing

- Building **multi-modal perception system** (RGB + thermal + audio) in **ROS2 + NVIDIA Isaac Sim** for navigation in low-visibility environments

Autonomous Mobile Robot

February 2025 – Ongoing

- Designed in **Creo Parametric** & deployed using **ROS2** and by implementing **EKF for state estimation (IMU + encoder fusion)** with **embedded (UART) communication pipeline** between **ESP32** and onboard compute for real- time closed loop control
- Diagnosed & resolved sensor drift and communication issues during real-world deployment | Drive Link

Knowledge Articulated Intelligent Robotic Assistance

October 2025 – November 2025

- Developed ROS-based autonomous robot with **navigation + face recognition (OpenCV)**; built **perception pipeline** and deployed system integrating **Jetson Nano, sensors, ESP32** for **real-time operation**
- Deployed on campus, successfully interacting with **100+ users and recognising 25+ individuals** | Drive Link

Facial Detection & Recognition Camera

March 2025

- Built **Raspberry Pi-based face recognition system with MySQL**; reduced false attendance entries by **~80%** in real world deployment